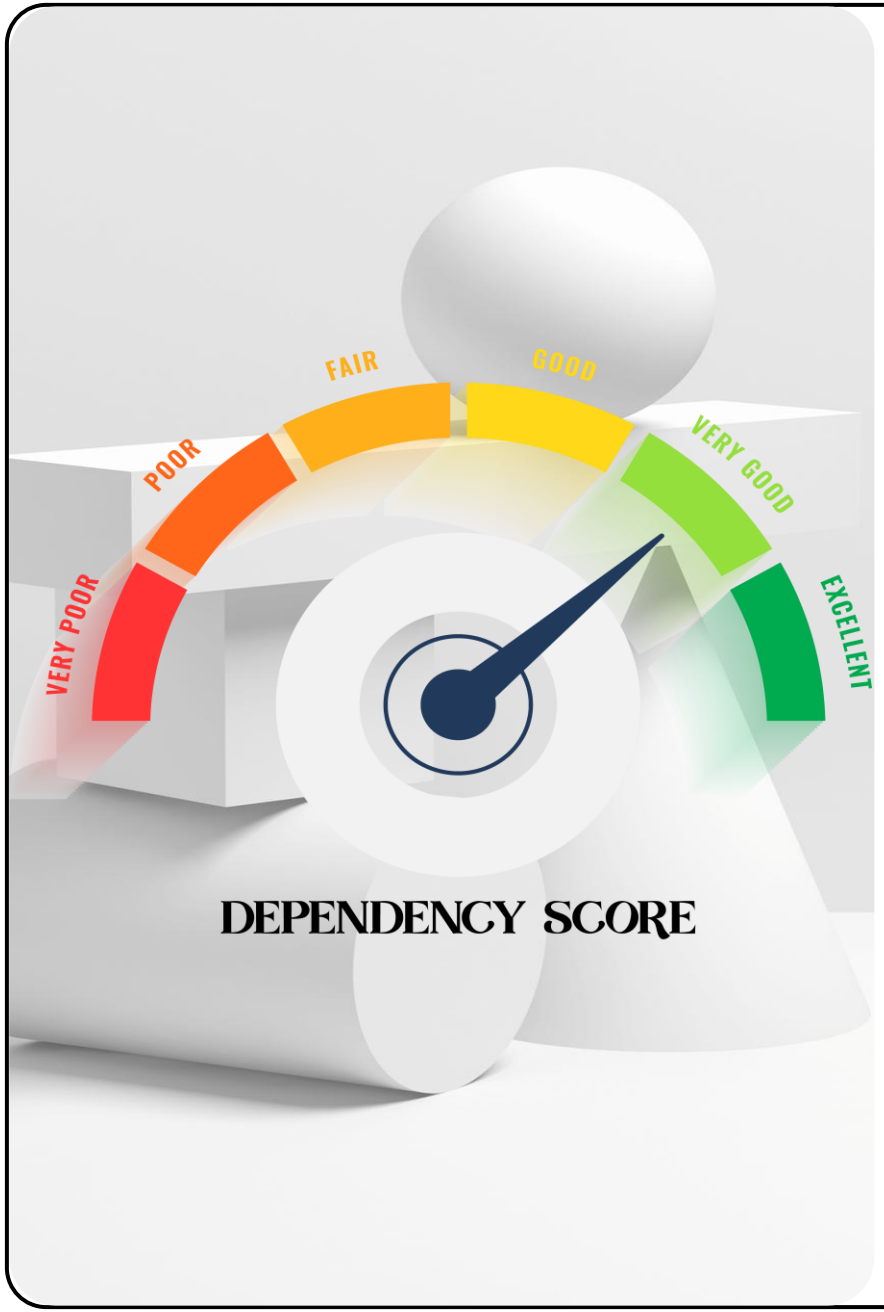
Data Analytics For International Business

ANALYZING US DEMAND SHOCKS AND INDUSTRIAL PRODUCTION IN EXPORT-ORIENTED ASIAN ECONOMIES



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INTRODUCTION

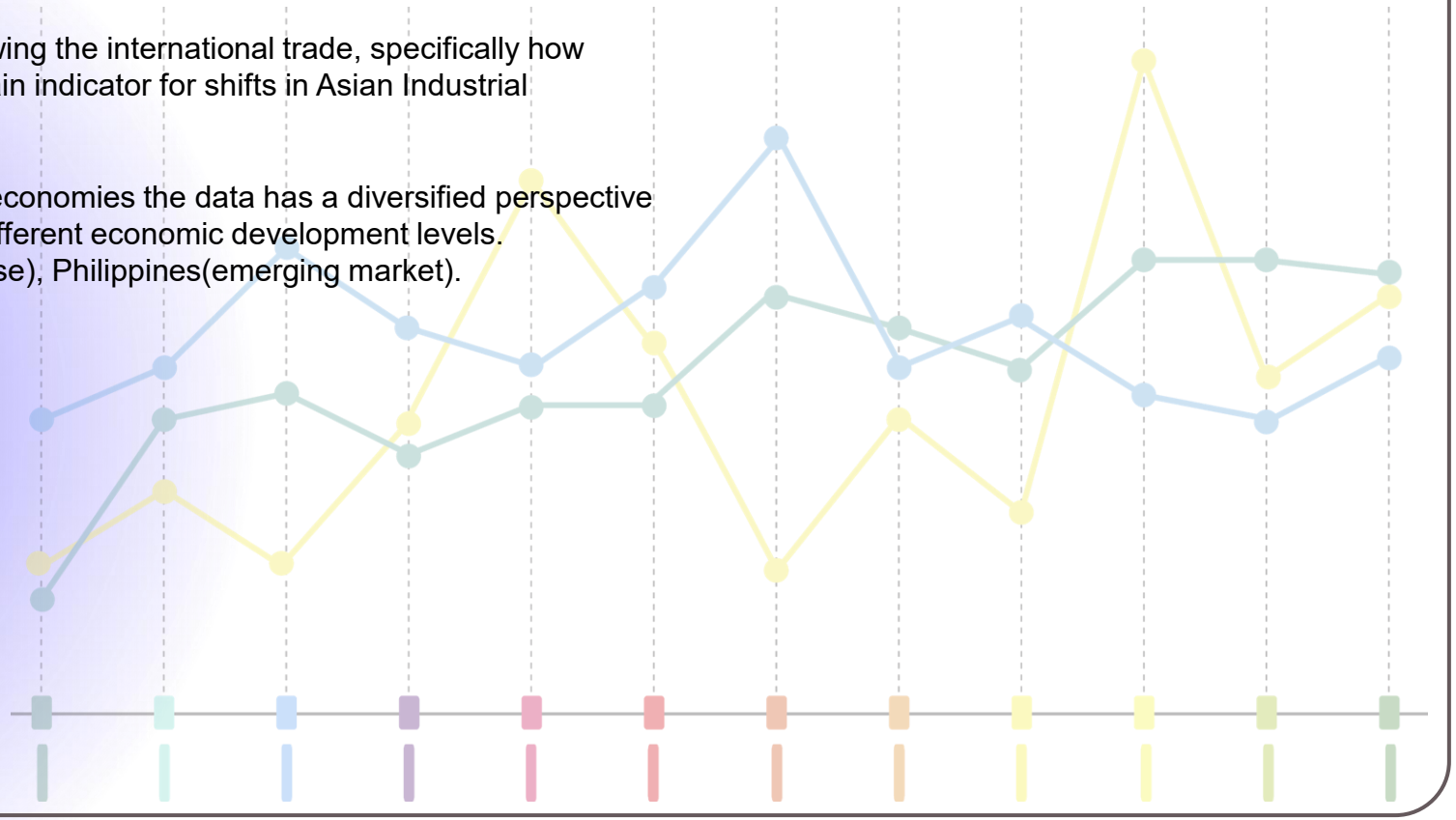
The interdependency between the Western and Eastern production forms the backbone of modern-day trade. The United States has always been the primary importer; it directly modulates the industrial output of manufacturing hubs. By applying time-series econometrics the technique move beyond simple correlation to quantify the precise lead time and magnitude of the ‘US Pull’ effect. By this study it is critical to understand supply chain resilience and predicting manufacturing cycles in a volatile global economy.

Assessing the Impact of US demand on Industrial Production: A detailed study of Singapore, Thailand, and the Philippines~ (1997-2017)

The Scope: This study analyses approx. 19-20 years period of macroeconomic data to determine the dependency of three key ASEAN(Association of Southeast Asian Nations) economies on US consumer behaviour.

The Core Mechanism: We are testing the ‘Pull Factor’ following the international trade, specifically how fluctuations in the US import index(log_usa_imp) act as a main indicator for shifts in Asian Industrial production(log_ind_prod_sa).

Comparative Dynamics: By including three different Asian economies the data has a diversified perspective and evaluates the US influence remains consistent across different economic development levels. Singapore(advance hub), Thailand(manufacturing powerhouse), Philippines(emerging market).



DEFINING THE CORE INQUIRY AND ANALYTICAL GOALS

The Primary Question: Does a statistically significant relationship exist between US import demand and the industrial production of Singapore, Thailand and the Philippines?

The goal here: To validate the “Export- Led growth” hypothesis

The Secondary Question: To what extent does a Global Value Chain(GVC) integration measure here as *export_gdp_ratio* intensifies the impact of US demand on the overall production and how much impact it really does to the system?

The goal here : To determine if higher trade dependency is creating higher vulnerability.

Analytical Objectives:

To normalize macroeconomic volatility using Log-Linear transformations.

To check and control the internal domestic factors such as Inflation(CPI) and Exchange Rates.

To quantify the Elasticity of the US-Asia trade link using multiple OLS regression.

EMPIRICAL FRAMEWORK AND VARIABLE IDENTIFICATION

- **Details**: Country specific data for Singapore(SGP), Thailand(THA) and the Philippines(PHL)
- **Dependent Variable***(Y): log_ind_prod_sa* : The log with seasonally adjusted industrial production index(Representing the manufacturing output).
- **Key Independent Variable***(X): log_usa_imp:* The natural log of US import demand(Representing the external demand pressure).

Control Variables:

- **cpi_yoy_nsa**: Consumer price index(domestic inflation).
- **Exchange_rate_vs_usd**: Local currency valuation against dollar(for trade competitiveness).
- **gvc_integration**: Export to GDP ratio(testing the trade participation and dependencies to external pull)

```
# EDA

df_dataset.info()
df
df_dataset.isnull().sum()
df_dataset.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 908 entries, 4 to 911
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   country               908 non-null    object
 1   countrycode           908 non-null    object
 2   year                  908 non-null    int64
 3   month                 908 non-null    int64
 4   cpi_yoy_nsa           908 non-null    float64
 5   USA_lag               908 non-null    float64
 6   exchnage_rate_vs_usd  908 non-null    float64
 7   ind_prod_const_sa     908 non-null    float64
 8   usa_imp_sa            908 non-null    float64
 9   Country Code          908 non-null    object
 10  Country Name          908 non-null    object
 11  gdp_pcap_ppp          908 non-null    object
 12  pop_total             908 non-null    object
 13  export_gdp_pct        908 non-null    float64
 14  log_ind_prod_sa       908 non-null    float64
 15  log_usa_imp           908 non-null    float64
dtypes: float64(8), int64(2), object(6)
memory usage: 152.9+ KB
```

	year	month	cpi_yoy_nsa	USA_lag	exchnage_rate_vs_usd	ind_prod_const_sa	usa_imp_sa	export_gdp_pct	log_ind_prod_sa	log_usa_imp
count	908.000000	908.000000	908.000000	908.000000	908.000000	9.080000e+02	908.000000	908.000000	908.000000	908.000000
mean	2008.039648	6.524229	2.596313	149817.076653	22.200371	7.455059e+09	150394.595708	98.169261	22.666602	11.880285
std	5.459541	3.442145	2.213753	38383.704374	20.333632	2.627273e+09	38281.498201	62.247842	0.371682	0.280014
min	1999.000000	1.000000	-4.272440	77781.000000	1.209087	2.692236e+09	79569.000000	26.177420	21.713638	11.261652
25%	2003.000000	4.000000	1.160243	106396.000000	2.698609	5.865551e+09	107040.000000	53.164962	22.492362	11.574923
50%	2008.000000	7.000000	2.373582	157913.000000	16.767580	6.586727e+09	158128.000000	69.730217	22.608322	11.969800
75%	2013.000000	10.000000	3.629721	186853.700834	41.959644	9.164940e+09	186963.000000	132.121211	22.938651	12.138081
max	2017.000000	12.000000	10.679436	203973.271221	56.330952	1.326830e+10	208877.825476	228.993771	23.308644	12.225744

MATHEMATICAL NORMALIZATION AND ELASTICITY INTERPRETATION

- **Addressing Skewness**: Raw macroeconomic data often exhibits “Positive Skew” (long right tails). Log transformation compresses these extreme values, pulling the distribution toward a Normal Bell curve.
- **Stabilizing Variance**: Macroeconomic variables often suffer from Heteroscedasticity. Logs help to stabilize the ‘noise’ and ensures the OLS model isn’t biased by high- volatility periods.
- **Economic Elasticity**: In a “Log-Log” model, the coefficients represent Elasticity, making and deriving an easier version to interpret.
- **Handling Scale** Difference: Log transformation allows the countries compare the countries of different economic size on a uniform percentage level.

```python
# Creating log transformations
df_dataset['log_ind_prod_sa']=np.log(df_dataset['ind_prod_const_sa'])
df_dataset['log_usa_imp']= np.log(df_dataset['USA_lag'])
```

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# EDA

df_dataset.info()
df
df_dataset.isnull().sum()
df_dataset.describe()
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<class 'pandas.core.frame.DataFrame'>
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 5   USA_lag               908 non-null    float64
 6   exchnage_rate_vs_usd  908 non-null    float64
 7   ind_prod_const_sa     908 non-null    float64
 8   usa_imp_sa            908 non-null    float64
 9   Country Code          908 non-null    object
 10  Country Name          908 non-null    object
 11  gdp_pcap_ppp          908 non-null    object
 12  pop_total             908 non-null    object
 13  export_gdp_pct        908 non-null    float64
 14  log_ind_prod_sa       908 non-null    float64
 15  log_usa_imp           908 non-null    float64
dtypes: float64(8), int64(2), object(6)
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```

Comparing Data Distributions: Raw vs. Log-Transformed

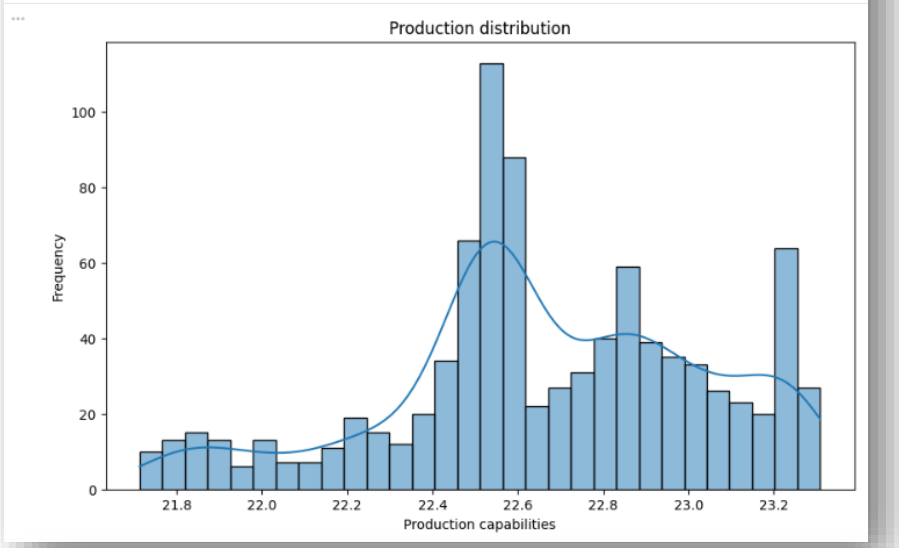
Raw Data: Initial histograms exhibit significant right skewness; this indicates that the mean is being pulled away from the median by extreme production spikes.

Log- Comparison effect: Applying natural logarithm compresses the wide gaps between high-value data points. This effectively pulls the long right tail back toward the center.

The Gaussian distribution: Post applying log it shows a symmetric bell curve. This satisfies that variable should follow normal distribution for the most reliable coefficient estimates as aligning with OLS logic.

The transition: Visually the graph moves from a skewed to a solid centralized mountain.

```
# Normality test to check the curve only in plot
plt.figure(figsize=(10,6))
sns.histplot(df_dataset['log_ind_prod_sa'], kde=True, bins=30)
plt.title('Production distribution')
plt.xlabel('Production capabilities')
plt.ylabel('Frequency')
plt.show()
```

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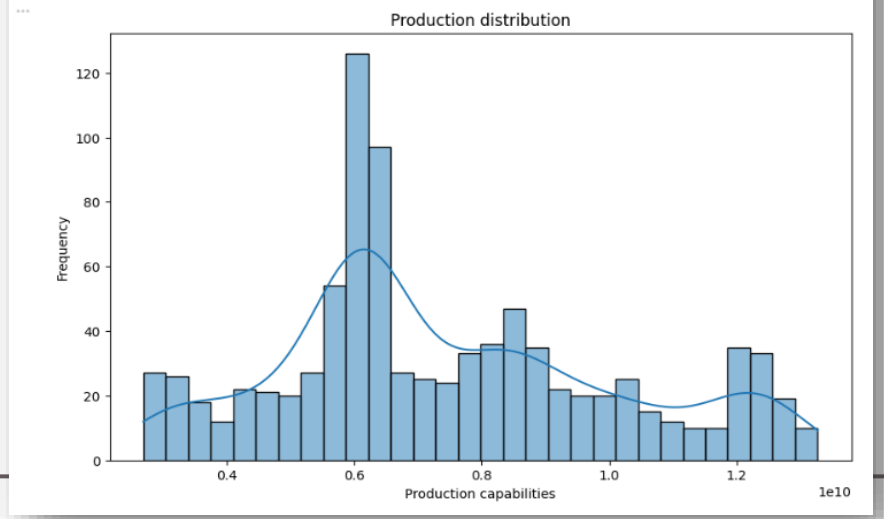


Addressing the “Philippines Problem”: From Skewed to Balanced Outliers

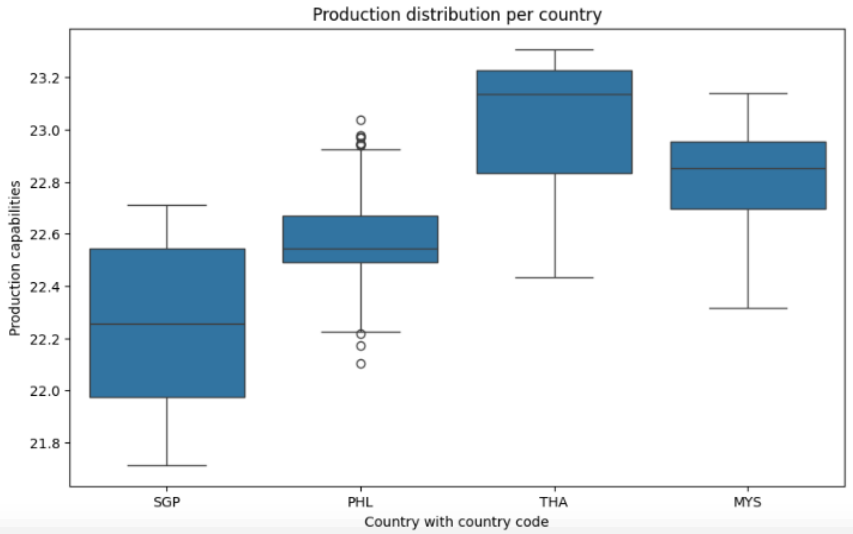
Raw Data: The initial boxplots for the Philippines(PHL) show a massive “Right-Tail” bias. Outliers are located on the top side indicating that few high- production periods were acting as a pull, making the average away from the true economic trend.

The Outlier Solution: Post log transformation, the boxplots reveal that outliers are now distributed more equally on both the top and bottom of Interquartile range(IQR).

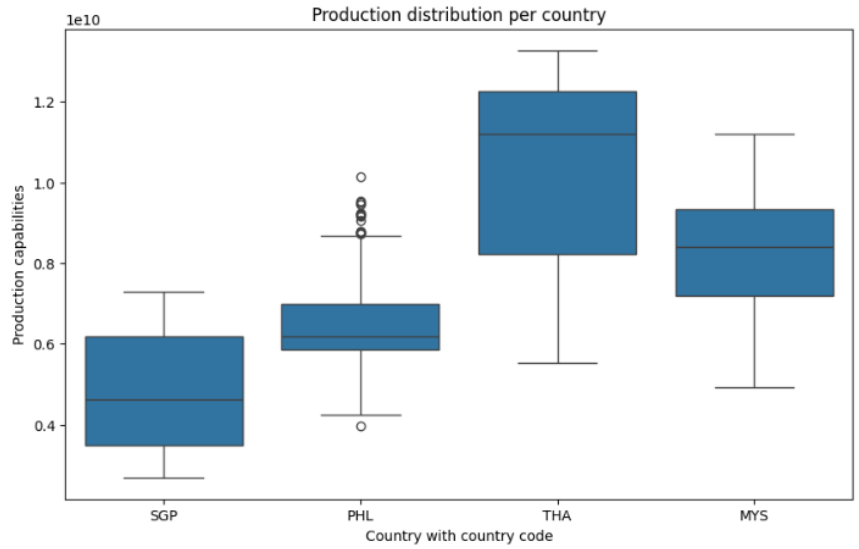
Logic and Stat: When outliers are balanced, they neutralize while calculating the OLS output. This results in an Unbiased Estimator, allowing the regression line to fit in the center of the data.

The Proof: The transition from one-sided(right tailed) to two-sided outliers is the visual sign that the data has a Symmetric normal distribution.

```python
# The box plot
plt.figure(figsize=(10,6))
sns.boxplot(x= 'countrycode', y= 'log_ind_prod_sa', data=df_dataset)
plt.title('Production distribution per country')
plt.xlabel('Country with country code')
plt.ylabel('Production capabilities')
plt.show()
```



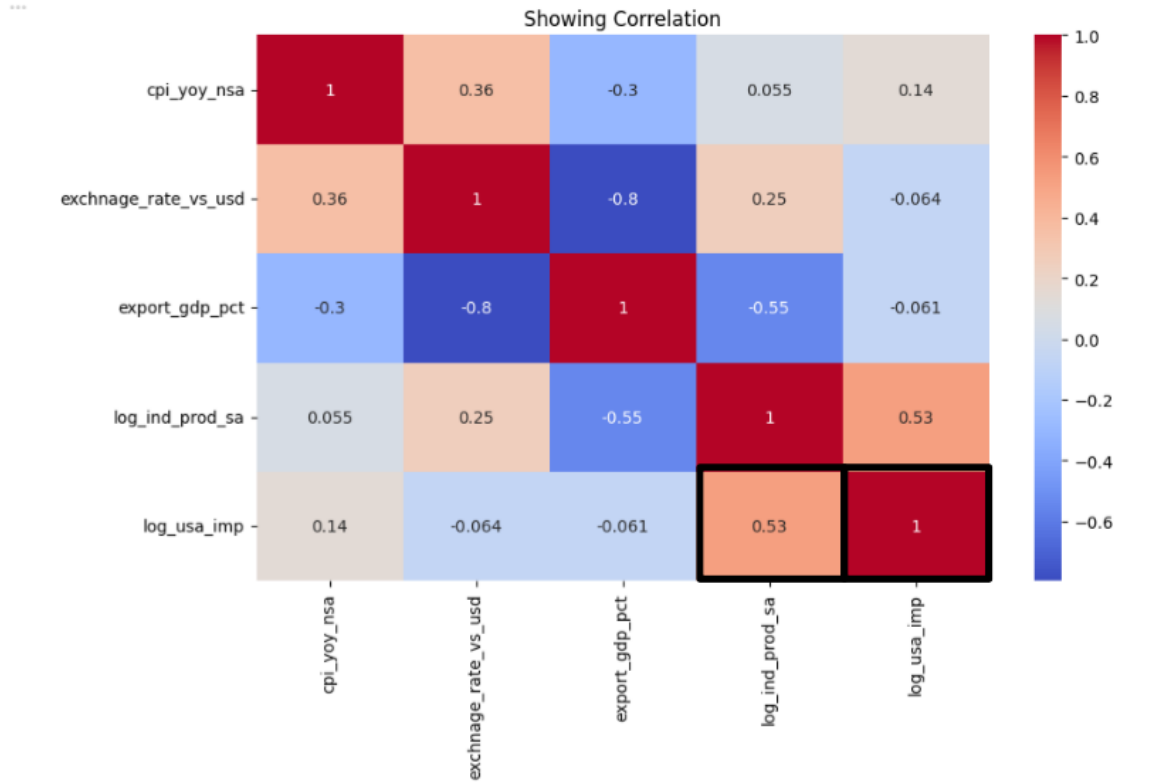

```python
# The box plot
plt.figure(figsize=(10,6))
sns.boxplot(x= 'countrycode', y= 'ind_prod_const_sa', data=df_dataset)
plt.title('Production distribution per country')
plt.xlabel('Country with country code')
plt.ylabel('Production capabilities')
plt.show()
```



QUANTIFYING THE LINEAR RELATIONSHIP BETWEEN TRADE VARIABLES

- **Strength of Association**: The heatmap provides the Pearson Correlation Coefficient, measuring the strength and direction of linear relationship between US demand and Asian Production.
- **Stable Correlation**: Post transformation the correlation remains robust . It indicates a mild to strong positive relationship where increases in US imports consistently align with increases in Asian Industrial output.
- **Identifying Multicollinearity**: This heatmap allows to check for high correlations between independent variables.
- **Warmth of the map**: Visually, the darker cells between *log_usa_imp* and *log_ind_prod_sa* confirms that the collinearity is functioning as hypothesized.

```python
# creating heatmap
plt.figure(figsize=(10,6))
sns.heatmap(correlation_log, annot= True, cmap='coolwarm')
plt.title('Showing Correlation')
plt.show()
```



Visualizing the Trend: Data Spreadability *vs.* Model Uniformity

The Trend: Scatter plots display the individual data points for each country plotted against US import demand. The red regression line represent the best fit calculated by the model.

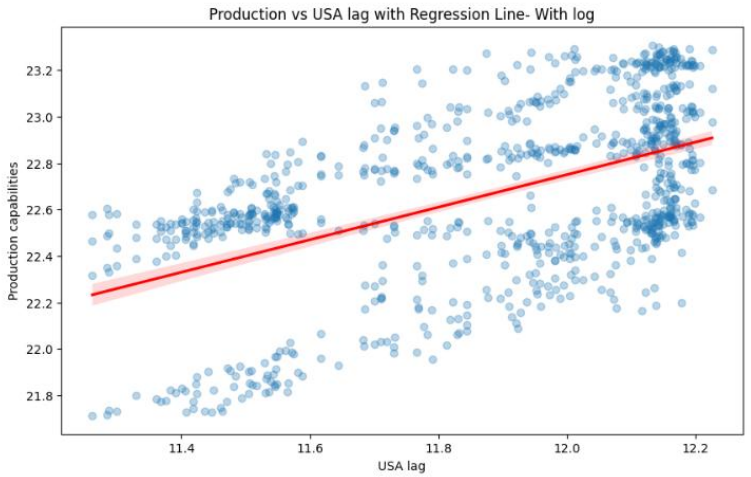
The Effect: In the raw plots, data points are highly dispersed(high variance) making it difficult for a single line to represent the trend accurately. Post- transformation, the points are more clustered much more tightly around the line.

Uniformity and steepness: The steepness of the slope in the log-transformed plot is a visual representation of elasticity.

Predictive Reliability: This tightening of the data points is the primary visual reason for the leap in R squared from 0.30 to 0.86. Lower dispersion=Higher predictive power.

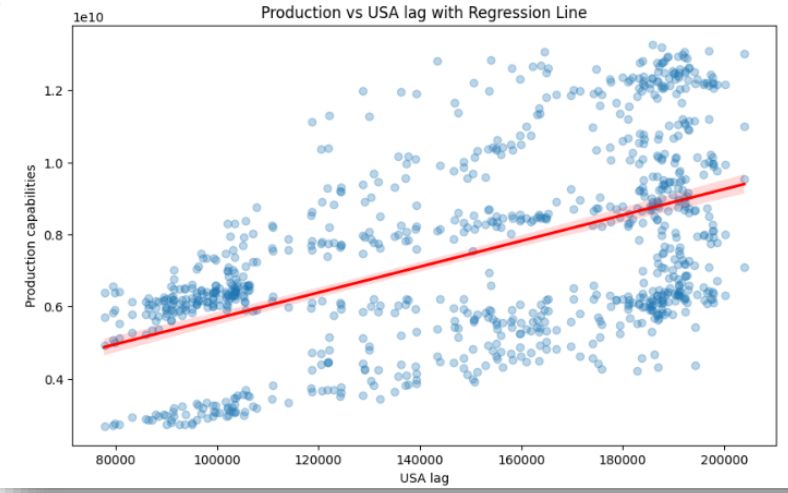
```python
# Creating a scatter plot to show the relationship between the production capacity of asia vs the USA lag post using log

plt.figure(figsize=(10,6))
sns.regplot(x= 'log_usa_imp', y='log_ind_prod_sa', data= df_dataset, scatter_kws={'alpha':0.3}, line_kws={'color': 'red'})
plt.title('Production vs USA lag with Regression Line- With log')
plt.xlabel('USA lag')
plt.ylabel('Production capabilities')
plt.show()
```




```python
# Creating a scatter plot to show the relationship between the production capacity of asia vs the USA lag

plt.figure(figsize=(10,6))
sns.regplot(x= 'USA_lag', y='ind_prod_const_sa', data= df_dataset, scatter_kws={'alpha':0.3}, line_kws={'color': 'red'})
plt.title('Production vs USA lag with Regression Line')
plt.xlabel('USA lag')
plt.ylabel('Production capabilities')
plt.show()
```



QUANTITATIVE EVIDENCE: EVALUATING MODEL ACCURACY AND EXPLANATORY POWER

- **Evolution of Accuracy**: Model A shows the R squared value is 0.30 showed that US demand alone explains only 30% of the movement. Model B by adding Fixed Effects and control variables, the adjusted R squared jumped to 0.86.
- **Global Significance**: The Prob(**F-stat**) of 0.000 confirms that the probability having zero effect is essentially zero. The model is statistically significant.
- **The Constant**: The intercept accounts for the baseline production level of the countries when all other factors are zero.
- **The Jump**: The jump from 30% to 86% proves that while US Demand is the Engine, the dependent variable is equally important for a precise prediction.

```
# run the model
model = sm_api.OLS(endog=y, exog=X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        log_ind_prod_sa   R-squared:                       0.866
Model:                            OLS   Adj. R-squared:                  0.864
Method:                 Least Squares   F-statistic:                     827.4
Date:                Tue, 17 Mar 2026   Prob (F-statistic):               0.00
Time:                        22:05:27   Log-Likelihood:                 521.62
No. Observations:                 908   AIC:                            -1027.
Df Residuals:                     900   BIC:                            -988.8
Df Model:                           7
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     14.4459      0.229     63.134      0.000      13.997      14.895
log_usa_imp                0.7061      0.018     38.377      0.000       0.670       0.742
gvc_integration        -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
cpi_yoy_nsa               -0.0063      0.003     -2.494      0.013      -0.011      -0.001
exchnage_rate_vs_usd      -0.0003      0.002     -0.221      0.825      -0.003       0.003
PHL                       -0.2054      0.070     -2.937      0.003      -0.343      -0.068
SGP                       -0.5667      0.040    -14.212      0.000      -0.645      -0.488
THA                        0.2074      0.052      4.007      0.000       0.106       0.309
==============================================================================
Omnibus:                        2.173   Durbin-Watson:                   2.064
Prob(Omnibus):                  0.337   Jarque-Bera (JB):                2.059
Skew:                           0.076   Prob(JB):                        0.357
Kurtosis:                       3.176   Cond. No.                     7.04e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.04e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

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                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
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gvc_integration       -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
cpi_yoy_nsa               -0.0063      0.003     -2.494      0.013      -0.011      -0.001
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PHL                       -0.2054      0.070     -2.937      0.003      -0.343      -0.068
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==============================================================================
Omnibus:                        2.173   Durbin-Watson:                   2.064
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Skew:                           0.076   Prob(JB):                        0.357
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```

QUANTIFYING THE "PULL EFFECT": FROM LOG-LINEAR COEFFICIENTS TO ECONOMIC REALITIES

- **The concept of Elasticity**: In a Log-Log model the coefficient for an independent variable represent the percentage change in the dependent variable.
- **The driver**: The coefficient for US import demand is 0.7061. It says 1% increase in US Import demand industrial production in these Asian markets increases by approximately 0.71%.. This is a high magnitude shows positive elasticity.
- **The Inflation Drag**: The coefficient is -0.0063. A 1% rise in domestic inflation leads to a minor decrease of 0.006% in industrial output. While the impact is small, the negative sign indicates that a rising costs act as a drag on production.
- **The currency Factor**: A coefficient of -0.0003 shows that the exchange rate volatility has a negligible, though slightly negative impact as compared to the demand pull in US.

STATISTICAL VALIDATION- HYPOTHESIS TEST

```python
# run the model
model = sm_api.OLS(endog=y, exog=X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        log_ind_prod_sa   R-squared:                       0.866
Model:                            OLS   Adj. R-squared:                  0.864
Method:                 Least Squares   F-statistic:                     827.4
Date:                Tue, 17 Mar 2026   Prob (F-statistic):               0.00
Time:                        22:05:27   Log-Likelihood:                 521.62
No. Observations:                 908   AIC:                            -1027.
Df Residuals:                     900   BIC:                            -988.8
Df Model:                           7
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     14.4459      0.229     63.134      0.000      13.997      14.895
log_usa_imp                0.7061      0.018     38.377      0.000       0.670       0.742
gvc_integration        -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
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```

- **The Null Hypothesis**(***Ho***): There is no significant relationship between US import demand and Asian Industrial Production.(*Coeff=0*)
- **Alternative Hypothesis**(***H1***): Yes, there is a significant relationship between US import demand and Asian Industrial Production.
- **The Stats**: *p-value* is 0.000, we reject the null hypothesis with 99.9% confidence.
- **Inference**: This confirms that the US market act as a “Primary Engine” for the manufacturing sectors of Singapore, Thailand and Philippines.
- **Dependencies**: This result suggest that these economies are majorly sensitive to US economic health.

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--------------------------------------------------------------------------------------------------
const                 14.4459      0.229     63.134      0.000      13.997      14.895
log_usa_imp            0.7061      0.018     38.377      0.000       0.670       0.742
gvc_integration    -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
cpi_yoy_nsa           -0.0063      0.003     -2.494      0.013      -0.011      -0.001
exchnage_rate_vs_usd  -0.0003      0.002     -0.221      0.825      -0.003       0.003
PHL                   -0.2054      0.070     -2.937      0.003      -0.343      -0.068
SGP                   -0.5667      0.040    -14.212      0.000      -0.645      -0.488
THA                    0.2074      0.052      4.007      0.000       0.106       0.309
==============================================================================
Omnibus:                        2.173   Durbin-Watson:                   2.064
Prob(Omnibus):                  0.337   Jarque-Bera (JB):                2.059
Skew:                           0.076   Prob(JB):                        0.357
Kurtosis:                       3.176   Cond. No.                     7.04e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.04e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

EVALUATION THE ROLE OF GLOBAL VALUE CHAIN (GVC) INTEGRATION

- **The Null Hypothesis**(***Ho***): GVC integration does not influence the US-Asia trade link(*Coefficient=0*)
- **The Result**: The *p-value* for *gvc_integration* is 0.924.
- **The Decision**: Since 0.924 is much greater than 0.05, we fail to reject the null hypothesis.
- **The Logic**: Absolute volume of US- import demand is much powerful that it overshadows the above values.
- **Insight**: This proves that these countries, the demand in US matters far more than the global value chain ratio.

DIAGNOSTIC TESTING:
VALIDATING THE BEST LINEAR UNBIASED ESTIMATOR STATUS

- **Autocorrelation** (**Durbin-Watson**): The result is 2.064. A value of 2.0 is ideal, our results proves there is no serial correlation, meaning the errors from one year are not affecting the next. Time-series is stable.
- **Normality of Residuals** (**Omnibus**/**Prob**): The *Prob(omnibus)* is 0.337, this is greater than 0.05 thus rejecting the null of normality. It proves a bell curve confirming the Log transformation was 100% successful.
- **Skewness**(**0.076**) and **Kurtosis**(**3.176**): It proves the model is perfect and the model is not being pulled or pushed by outliers, concluding as balanced and reliable model.

```python
# run the model
model = sm_api.OLS(endog=y, exog=X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        log_ind_prod_sa   R-squared:                       0.866
Model:                            OLS   Adj. R-squared:                  0.864
Method:                 Least Squares   F-statistic:                     827.4
Date:                Tue, 17 Mar 2026   Prob (F-statistic):               0.00
Time:                        22:05:27   Log-Likelihood:                 521.62
No. Observations:                 908   AIC:                            -1027.
Df Residuals:                     900   BIC:                            -988.8
Df Model:                           7
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                14.4459      0.229     63.134      0.000      13.997      14.895
log_usa_imp           0.7061      0.018     38.377      0.000       0.670       0.742
gvc_integration  -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
cpi_yoy_nsa          -0.0063      0.003     -2.494      0.013      -0.011      -0.001
exchnage_rate_vs_usd -0.0003      0.002     -0.221      0.825      -0.003       0.003
PHL                  -0.2054      0.070     -2.937      0.003      -0.343      -0.068
SGP                  -0.5667      0.040    -14.212      0.000      -0.645      -0.488
THA                   0.2074      0.052      4.007      0.000       0.106       0.309
==============================================================================
Omnibus:                        2.173   Durbin-Watson:                   2.064
Prob(Omnibus):                  0.337   Jarque-Bera (JB):                2.059
Skew:                           0.076   Prob(JB):                        0.357
Kurtosis:                       3.176   Cond. No.                     7.04e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.04e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

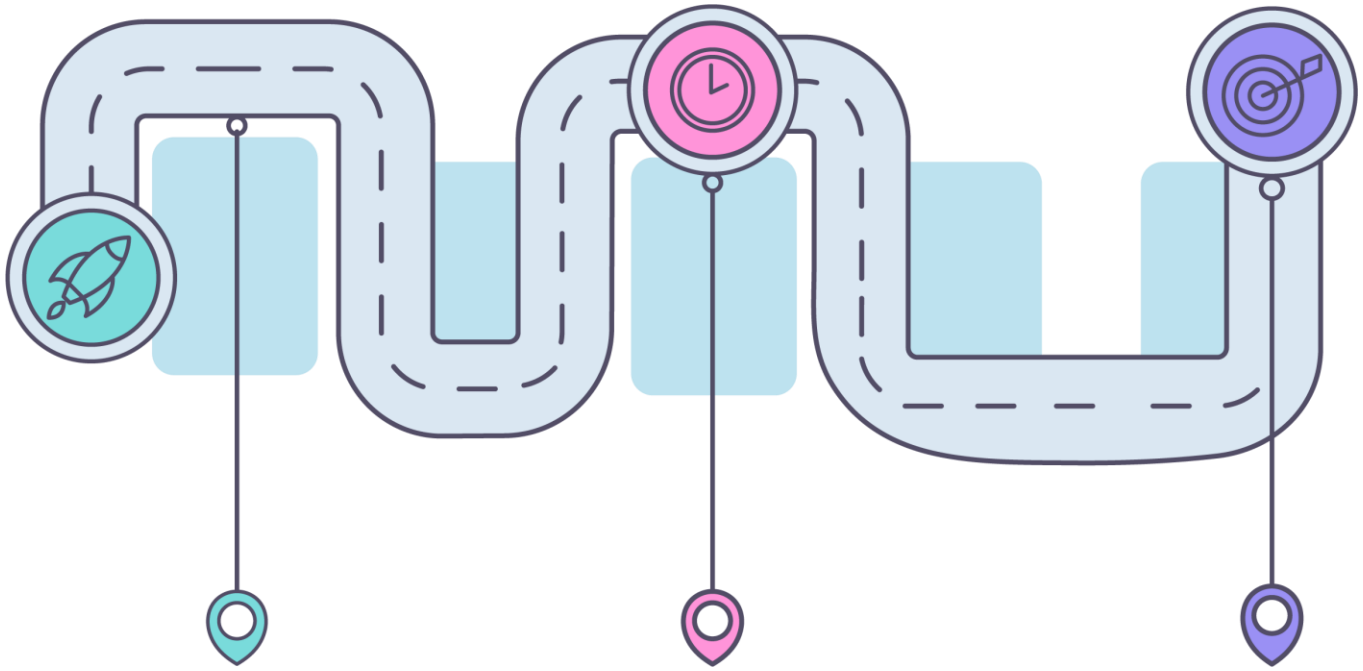
STRATEGIC SYNTHESIS: NAVIGATING EXTERNAL DEPENDENCY

Validation: The research confirms that US import demand is the single most powerful predictor of industrial production for the target ASEAN markets.(0.71).

Analysis: The 86% model accuracy proves these nations are aligned with the US business cycle. A recession in US will bring guaranteed downfall to production in Asia.

Strategy 1: To mitigate 0.71 elasticity dependency, policymakers must aggressively take market diversification.

Strategy 2: CPI negatively impacts output; the central bank should focus on stabilizing market. Maintaining a low CPI ensures the time frame in meeting the external demand should be competitive and predictable.



Start points Action Steps End points

Outlook: From mass production to tech-base process which reduces the dependency US goods, as the GDP rises it leads to push the countries towards innovations. The goal is to create diversified space, US will try to spread out the import structure. Asian markets will focus to become tech hubs

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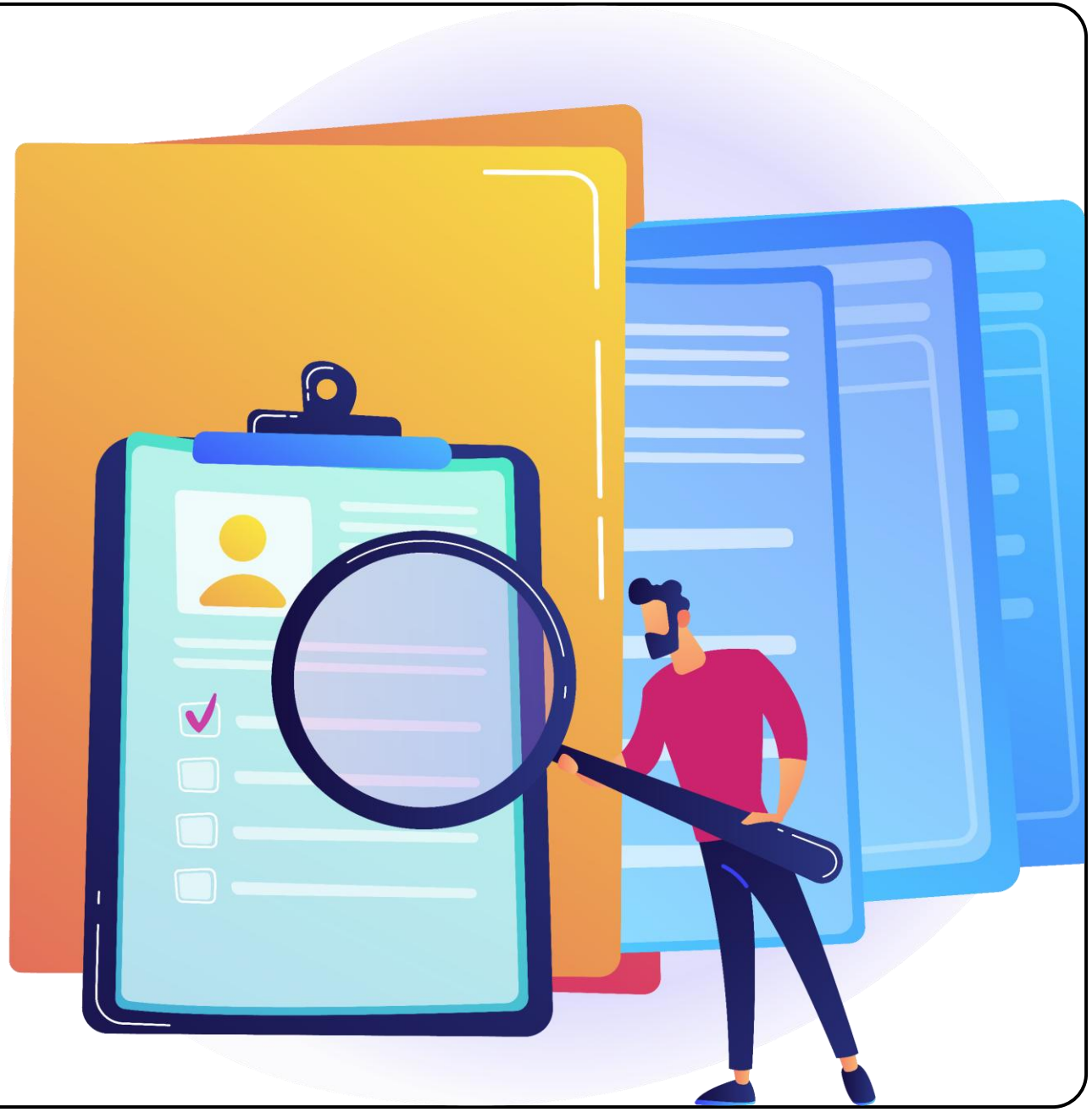

CONTD...

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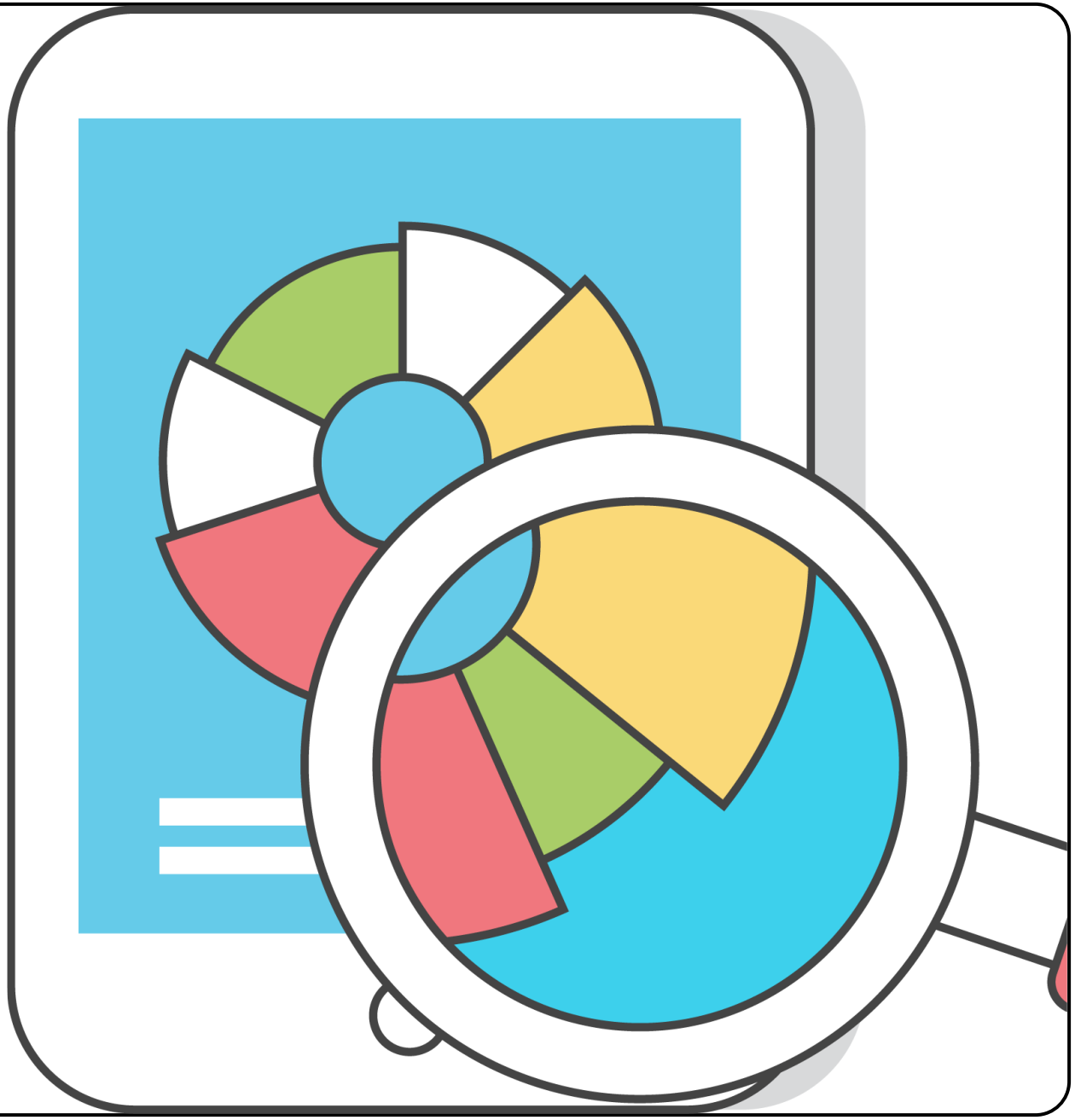


THANK YOU

Q & A

Panel discussion session

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#main excel
df=pd.read_excel('asia-industry.xlsx')
df=df[(df['year']>=1999) & (df['year']<=2017)].copy()
```

```
#loading other files
df_export=pd.read_excel('export % of GDP.xlsx')
df_long= df_export.melt(id_vars=['Country Code', 'Country Name', 'Series Name', 'Variable Name'],
                        var_name='years_raw',
                        value_name='export_gdp_pct')

df_long
```

	Country Code	Country Name	Series Name	Variable Name	years_raw	export_gdp_pct
0	MYS	Malaysia	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	115.743725
1	PHL	Philippines	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	37.806868
2	SGP	Singapore	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	166.787571
3	THA	Thailand	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	57.937303
4	MYS	Malaysia	Exports of goods and services (% of GDP)	exp_per_GDP	1999 [YR1999]	121.311394
...	...	...	...	...	...	...
79	THA	Thailand	Exports of goods and services (% of GDP)	exp_per_GDP	2017 [YR2017]	66.672832
80	MYS	Malaysia	Exports of goods and services (% of GDP)	exp_per_GDP	2018 [YR2018]	68.554971
81	PHL	Philippines	Exports of goods and services (% of GDP)	exp_per_GDP	2018 [YR2018]	30.213605
82	SGP	Singapore	Exports of goods and services (% of GDP)	exp_per_GDP	2018 [YR2018]	178.198251
83	THA	Thailand	Exports of goods and services (% of GDP)	exp_per_GDP	2018 [YR2018]	64.838097

84 rows × 6 columns

```
# # .melt
# id_verse
# var_name
# value_name
#b4
# Country Code,1990 [YR1990],1991 [YR1991],1992 [YR1992]
# AFG,10.5,11.2,10.8
# ALB,15.2,14.8,16.1
#a4
# Country Code,year_raw,exports_gdp_pct
# AFG,1990 [YR1990],10.5
# AFG,1991 [YR1991],11.2
# AFG,1992 [YR1992],10.8
# ALB,1990 [YR1990],15.2
# ALB,1991 [YR1991],14.8
# ALB,1992 [YR1992],16.1
```

```
# The 'years_raw' column does not contain only numerical year, so we need to extract it.
df_long['year_clean'] = df_long['years_raw'].str.extract('(\d{4})').astype(int)

# Display the updated DataFrame (optional)
df_long.head()
```

```
<>:2: SyntaxWarning: invalid escape sequence '\d'
<>:2: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipykernel_202/1752635651.py:2: SyntaxWarning: invalid escape sequence '\d'
  df_long['year_clean'] = df_long['years_raw'].str.extract('(\d{4})').astype(int)
```

	Country Code	Country Name	Series Name	Variable Name	years_raw	export_gdp_pct	year_clean
0	MYS	Malaysia	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	115.743725	1998
1	PHL	Philippines	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	37.806868	1998
2	SGP	Singapore	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	166.787571	1998
3	THA	Thailand	Exports of goods and services (% of GDP)	exp_per_GDP	1998 [YR1998]	57.937303	1998
4	MYS	Malaysia	Exports of goods and services (% of GDP)	exp_per_GDP	1999 [YR1999]	121.311394	1999

```
# merging

df_data=pd.merge(df,df_long,
                 left_on=['countrycode','year'],
                 right_on=['Country Code','year_clean'],
                 how='left'
                 )
```

```
# df_data = pd.merge(df, df_long,
#                  left_on=['countrycode','year'],
#                  right_on=['Country Code','year_clean'],
#                  how='left')


# df_pop=pd.read_excel('total population.xlsx')
# df_popp_processed = df_pop.melt(id_vars=['Country Code'],
#                              value_vars=[col for col in df_pop.columns if isinstance(col, (int, float)) or 'YR' in str(col)],
#                              var_name='Years_raw_pop',
#                              value_name='population_total')

# df_popp_processed['year'] = df_popp_processed['Years_raw_pop'].astype(str).str.extract(r'(\d{4})').astype(int)
# df_popp_processed.drop(columns=['Years_raw_pop'], inplace=True)
# df_popp_processed.rename(columns={'Country Code': 'countrycode'}, inplace=True)


# # GDP
# df_gdp=pd.read_excel('GDP per Capital PPP (2021 constant international $).xlsx')
# df_gdpm_processed = df_gdp.melt(id_vars=['Country Code'],
#                              value_vars=[col for col in df_gdp.columns if isinstance(col, (int, float)) or 'YR' in str(col)],
#                              var_name='Years_raw_gdp',
#                              value_name='gdp_per_capita')

# df_gdpm_processed['year'] = df_gdpm_processed['Years_raw_gdp'].astype(str).str.extract(r'(\d{4})').astype(int)
# df_gdpm_processed.drop(columns=['Years_raw_gdp'], inplace=True)

# df_gdpm_processed['gdp_per_capita'] = pd.to_numeric(df_gdpm_processed['gdp_per_capita'], errors='coerce').astype('Int64')

# df_gdpm_processed.rename(columns={'Country Code': 'countrycode'}, inplace=True)


# #Merge population into our growing df_data
# df_data = pd.merge(df_data,
#                   df_popp_processed,
#                   on=['countrycode', 'year'],
#                   how='left'
#                  )


# #Merge GDP
# df_data = pd.merge(df_data,
#                   df_gdpm_processed,
#                   on=['countrycode', 'year'],
#                   how='left'
#                  )
```

```
df_pop = pd.read_excel('total population.xlsx')

# Melt
df_pop_long = df_pop.melt(id_vars=['Country Code'], var_name='year_raw', value_name='pop_total')


df_pop_long['year_clean'] = df_pop_long['year_raw'].str.extract('(\d{4})')
```

```
df_pop_long = df_pop_long.dropna(subset=['year_clean'])

df_pop_long['year'] = df_pop_long['year_clean'].astype(int)



df_gdp = pd.read_excel('GDP per Capital PPP (2021 constant international $).xlsx')

df_gdp_long = df_gdp.melt(id_vars=['Country Code'], var_name='year_raw', value_name='gdp_pcap_ppp')

df_gdp_long['year_clean'] = df_gdp_long['year_raw'].str.extract('(\d{4})')

df_gdp_long = df_gdp_long.dropna(subset=['year_clean'])

df_gdp_long['year'] = df_gdp_long['year_clean'].astype(int)
```

Show hidden output

```
# Merge population into our growing df_data
df_pop_long = df_pop_long.rename(columns={'Country Code': 'countrycode'})

df_dataF = pd.merge(df_data,
                 df_pop_long,
                 on=['countrycode', 'year'],
                 how='left'
                )


# Merge GDP
df_gdp_long = df_gdp_long.rename(columns={'Country Code': 'countrycode'})

df_data_final = pd.merge(df_dataF,
                 df_gdp_long,
                 on=['countrycode', 'year'],
                 how='left'
                )
```

```
df_data_final
```

	country	countrycode	year	month	cpi_yoy_nsa	exchnage_rate_vs_usd	ind_prod_const_sa	usa_imp_sa	Country Code	Country Name	...	Variable Name	years_raw	export_gdp_pct	year_clean_x	year_raw_x	pop_total	year_clean_y	year_raw_y	gdp_pcap_ppp	year_clean
0	Malaysia	MYS	1999	1	5.181375	3.800000	4.577605e+09	77781.000000	MYS	Malaysia	...	exp_per_GDP	1999 [YR1999]	121.311394	1999	1999	22435531	1999	1999	17428.625656	1999
1	Singapore	SGP	1999	1	-0.661324	1.678571	2.589487e+09	77781.000000	SGP	Singapore	...	exp_per_GDP	1999 [YR1999]	176.745013	1999	1999	3958723	1999	1999	63746.479223	1999
2	Philippines	PHL	1999	1	10.555900	38.294286	6.522064e+09	77781.000000	PHL	Philippines	...	exp_per_GDP	1999 [YR1999]	38.411159	1999	1999	77753257	1999	1999	4650.759269	1999
3	Thailand	THA	1999	1	3.273102	36.593810	6.098279e+09	77781.000000	THA	Thailand	...	exp_per_GDP	1999 [YR1999]	56.438543	1999	1999	62299043	1999	1999	11253.387496	1999
4	Singapore	SGP	1999	2	-0.987156	1.699600	2.692236e+09	79842.000000	SGP	Singapore	...	exp_per_GDP	1999 [YR1999]	176.745013	1999	1999	3958723	1999	1999	63746.479223	1999
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
907	Singapore	SGP	2017	11	0.612723	1.355182	7.093773e+09	203973.271221	SGP	Singapore	...	exp_per_GDP	2017 [YR2017]	171.741626	2017	2017	5612253	2017	2017	116471.820765	2017
908	Malaysia	MYS	2017	12	3.588902	4.067065	1.100520e+10	208877.825476	MYS	Malaysia	...	exp_per_GDP	2017 [YR2017]	70.011743	2017	2017	32355644	2017	2017	29551.54576	2017
909	Thailand	THA	2017	12	0.761127	32.554595	1.300725e+10	208877.825476	THA	Thailand	...	exp_per_GDP	2017 [YR2017]	66.672832	2017	2017	71160187	2017	2017	20094.677876	2017
910	Philippines	PHL	2017	12	3.074009	50.478905	9.533997e+09	208877.825476	PHL	Philippines	...	exp_per_GDP	2017 [YR2017]	29.552291	2017	2017	108119693	2017	2017	8584.136227	2017
911	Singapore	SGP	2017	12	0.356776	1.345633	7.104364e+09	208877.825476	SGP	Singapore	...	exp_per_GDP	2017 [YR2017]	171.741626	2017	2017	5612253	2017	2017	116471.820765	2017

912 rows × 21 columns

```
df_data_final.to_csv('df_data_final', index=False)
print("Merged dataset saved to 'df_merged.csv'")
```

```
Merged dataset saved to 'df_merged.csv'
```

```
df_data_final.to_excel('df_data_final.xlsx', index=False)
print("Merged dataset saved to 'df_data_final.xlsx'")
```

```
Merged dataset saved to 'df_data_final.xlsx'
```

```
# Creating LAG space
df_data_final['USA_lag']= df_data_final.groupby('countrycode')['usa_imp_sa'].shift(1)
```

```
# now cleanning
cols= ['country', 'countrycode', 'year','month',
       'cpi_yoy_nsa','USA_lag', 'exchnage_rate_vs_usd', 'ind_prod_const_sa',    'usa_imp_sa', 'Country Code',   'Country Name', 'gdp_pcap_ppp','pop_total','export_gdp_pct']

df_dataset= df_data_final[cols]
df_dataset= df_dataset.dropna(subset=['USA_lag'])
df_dataset.head()
```

	country	countrycode	year	month	cpi_yoy_nsa	USA_lag	exchnage_rate_vs_usd	ind_prod_const_sa	usa_imp_sa	Country Code	Country Name	gdp_pcap_ppp	pop_total	export_gdp_pct
4	Singapore	SGP	1999	2	-0.987156	77781.0	1.699600	2.692236e+09	79842.0	SGP	Singapore	63746.479223	3958723	176.745013
5	Philippines	PHL	1999	2	9.055043	77781.0	38.811250	6.383731e+09	79842.0	PHL	Philippines	4650.759269	77753257	38.411159
6	Thailand	THA	1999	2	2.488452	77781.0	37.060500	5.710419e+09	79842.0	THA	Thailand	11253.387496	62299043	56.438543
7	Malaysia	MYS	1999	2	3.850393	77781.0	3.800000	4.915195e+09	79842.0	MYS	Malaysia	17428.625656	22435531	121.311394
8	Singapore	SGP	1999	3	-0.519663	79842.0	1.729435	2.751341e+09	79569.0	SGP	Singapore	63746.479223	3958723	176.745013

```
# Creating log transformations
df_dataset['log_ind_prod_sa']=np.log(df_dataset['ind_prod_const_sa'])
df_dataset['log_usa_imp']= np.log(df_dataset['USA_lag'])
```

```
# EDA

df_dataset.info()
df
df_dataset.isnull().sum()
df_dataset.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 908 entries, 4 to 911
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   country               908 non-null    object
 1   countrycode           908 non-null    object
 2   year                  908 non-null    int64
 3   month                 908 non-null    int64
 4   cpi_yoy_nsa           908 non-null    float64
 5   USA_lag               908 non-null    float64
 6   exchnage_rate_vs_usd  908 non-null    float64
 7   ind_prod_const_sa     908 non-null    float64
 8   usa_imp_sa            908 non-null    float64
 9   Country Code          908 non-null    object
 10  Country Name          908 non-null    object
 11  gdp_pcap_ppp          908 non-null    object
 12  pop_total             908 non-null    object
 13  export_gdp_pct        908 non-null    float64
 14  log_ind_prod_sa       908 non-null    float64
 15  log_usa_imp           908 non-null    float64
dtypes: float64(8), int64(2), object(6)
memory usage: 152.9+ KB
```

	year	month	cpi_yoy_nsa	USA_lag	exchnage_rate_vs_usd	ind_prod_const_sa	usa_imp_sa	export_gdp_pct	log_ind_prod_sa	log_usa_imp
count	908.000000	908.000000	908.000000	908.000000	908.000000	9.080000e+02	908.000000	908.000000	908.000000	908.000000
mean	2008.039648	6.524229	2.596313	149817.076653	22.200371	7.455059e+09	150394.595708	98.169261	22.666602	11.880285
std	5.459541	3.442145	2.213753	38383.704374	20.333632	2.627273e+09	38281.498201	62.247842	0.371682	0.280014
min	1999.000000	1.000000	-4.272440	77781.000000	1.209087	2.692236e+09	79569.000000	26.177420	21.713638	11.261652
25%	2003.000000	4.000000	1.160243	106396.000000	2.698609	5.865551e+09	107040.000000	53.164962	22.492362	11.574923
50%	2008.000000	7.000000	2.373582	157913.000000	16.767580	6.586727e+09	158128.000000	69.730217	22.608322	11.969800
75%	2013.000000	10.000000	3.629721	186853.700834	41.959644	9.164940e+09	186963.000000	132.121211	22.938651	12.138081
max	2017.000000	12.000000	10.679436	203973.271221	56.330952	1.326830e+10	208877.825476	228.993771	23.308644	12.225744

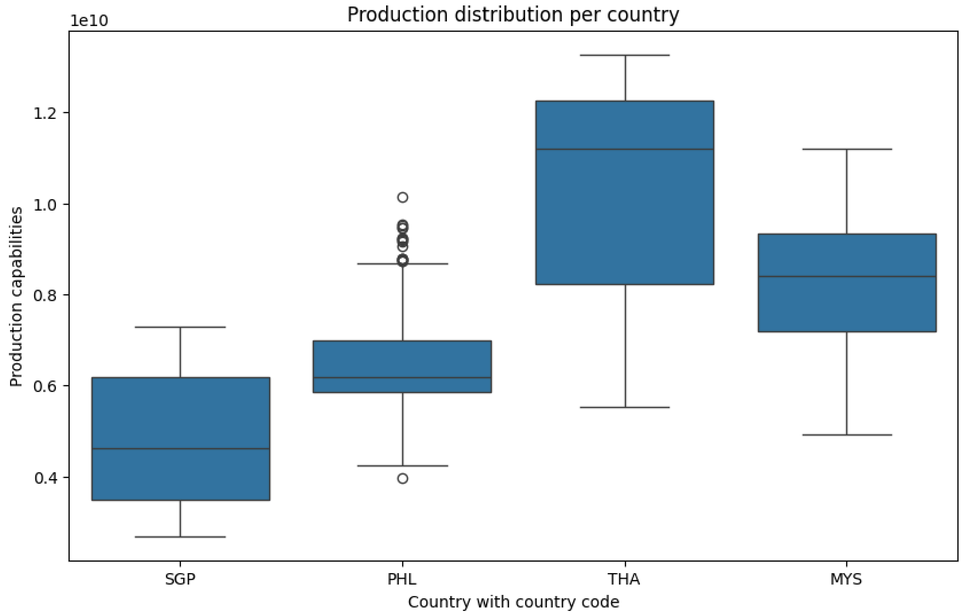
```
# Adding a detailed summary here
eda_summary= df_dataset.describe().T
eda_summary
```

	count	mean	std	min	25%	50%	75%	max
year	908.0	2.008040e+03	5.459541e+00	1.999000e+03	2.003000e+03	2.008000e+03	2.013000e+03	2.017000e+03
month	908.0	6.524229e+00	3.442145e+00	1.000000e+00	4.000000e+00	7.000000e+00	1.000000e+01	1.200000e+01
cpi_yoy_nsa	908.0	2.596313e+00	2.213753e+00	-4.272440e+00	1.160243e+00	2.373582e+00	3.629721e+00	1.067944e+01
USA_lag	908.0	1.498171e+05	3.838370e+04	7.778100e+04	1.063960e+05	1.579130e+05	1.868537e+05	2.039733e+05
exchnage_rate_vs_usd	908.0	2.220037e+01	2.033363e+01	1.209087e+00	2.698609e+00	1.676758e+01	4.195964e+01	5.633095e+01
ind_prod_const_sa	908.0	7.455059e+09	2.627273e+09	2.692236e+09	5.865551e+09	6.586727e+09	9.164940e+09	1.326830e+10
usa_imp_sa	908.0	1.503946e+05	3.828150e+04	7.956900e+04	1.070400e+05	1.581280e+05	1.869630e+05	2.088778e+05
export_gdp_pct	908.0	9.816926e+01	6.224784e+01	2.617742e+01	5.316496e+01	6.973022e+01	1.321212e+02	2.289938e+02
log_ind_prod_sa	908.0	2.266660e+01	3.716818e-01	2.171364e+01	2.249236e+01	2.260832e+01	2.293865e+01	2.330864e+01
log_usa_imp	908.0	1.188029e+01	2.800143e-01	1.126165e+01	1.157492e+01	1.196980e+01	1.213808e+01	1.222574e+01

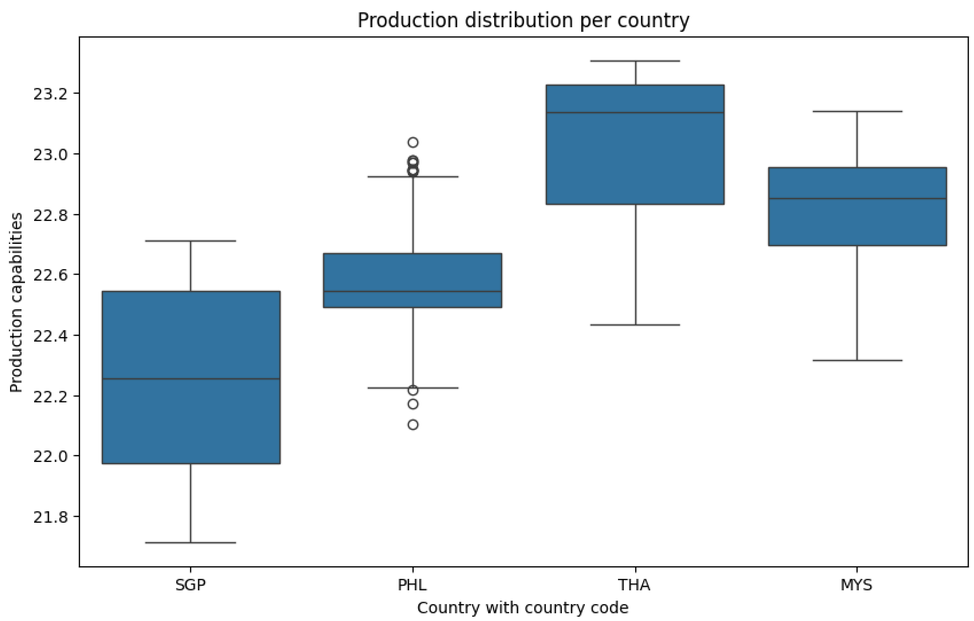
```
# Checking the mean and median for verification

# Convert 'gdp_pcap_ppp' and 'pop_total' to numeric, coercing errors
df_dataset['gdp_pcap_ppp'] = pd.to_numeric(df_dataset['gdp_pcap_ppp'], errors='coerce')
df_dataset['pop_total'] = pd.to_numeric(df_dataset['pop_total'], errors='coerce')


# Checking median
eda_summary['median'] = df_dataset.median(numeric_only=True)
# Checking variance
eda_summary['variance']= df_dataset.var(numeric_only= True)

#taking out specifics
print(eda_summary[['mean', 'median', 'std', 'variance', 'min', 'max']])
```

```
                             mean        median           std      variance  \
year                 2.008040e+03  2.008000e+03  5.459541e+00  2.980659e+01
month                6.524229e+00  7.000000e+00  3.442145e+00  1.184836e+01
cpi_yoy_nsa          2.596313e+00  2.373582e+00  2.213753e+00  4.900702e+00
USA_lag              1.498171e+05  1.579130e+05  3.838370e+04  1.473309e+09
exchnage_rate_vs_usd  2.220037e+01  1.676758e+01  2.033363e+01  4.134566e+02
ind_prod_const_sa    7.455059e+09  6.586727e+09  2.627273e+09  6.902564e+18
usa_imp_sa           1.503946e+05  1.581280e+05  3.828150e+04  1.465473e+09
export_gdp_pct       9.816926e+01  6.973022e+01  6.224784e+01  3.874794e+03
log_ind_prod_sa      2.266660e+01  2.260832e+01  3.716818e-01  1.381474e-01
log_usa_imp          1.188029e+01  1.196980e+01  2.800143e-01  7.840800e-02

                              min           max
year                 1.999000e+03  2.017000e+03
month                1.000000e+00  1.200000e+01
cpi_yoy_nsa         -4.272440e+00  1.067944e+01
USA_lag              7.778100e+04  2.039733e+05
exchnage_rate_vs_usd  1.209087e+00  5.633095e+01
ind_prod_const_sa    2.692236e+09  1.326830e+10
usa_imp_sa           7.956900e+04  2.088778e+05
export_gdp_pct       2.617742e+01  2.289938e+02
log_ind_prod_sa      2.171364e+01  2.330864e+01
log_usa_imp          1.126165e+01  1.222574e+01
```

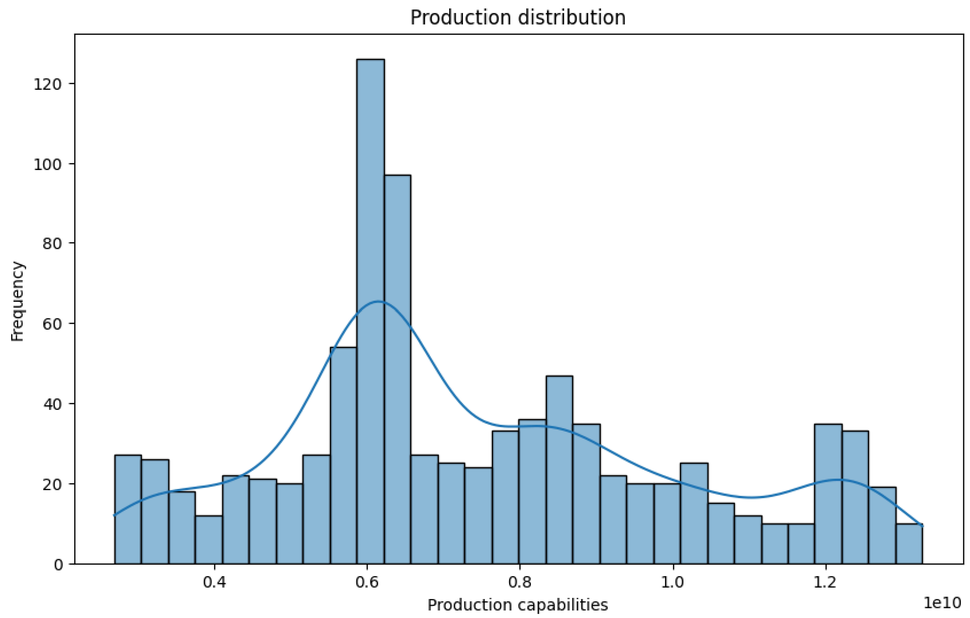
```
# The box plot
plt.figure(figsize=(10,6))
sns.boxplot(x= 'countrycode', y= 'ind_prod_const_sa', data=df_dataset)
plt.title('Production distribution per country')
plt.xlabel('Country with country code')
plt.ylabel('Production capabilities')
plt.show()
```



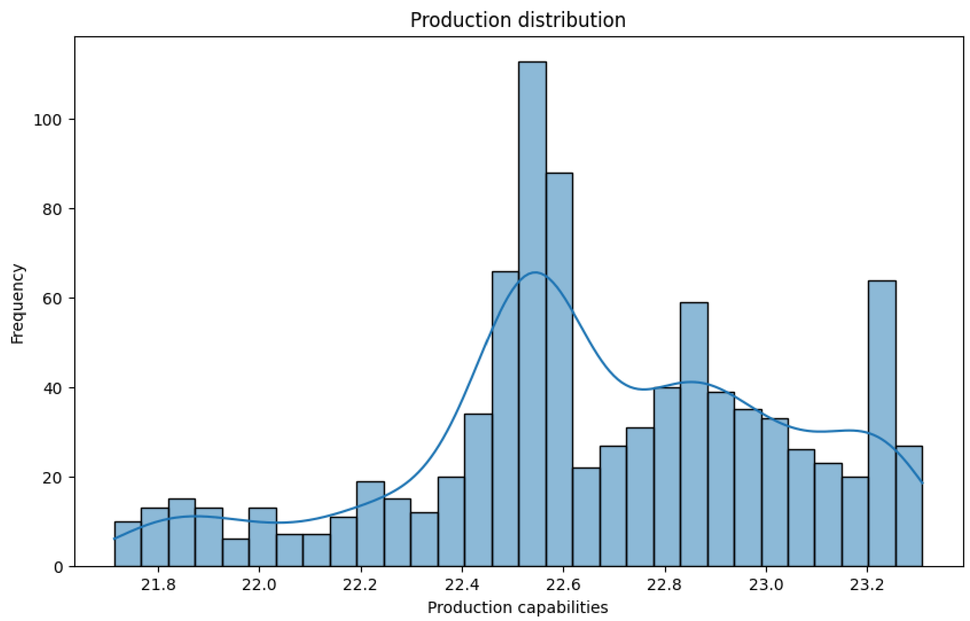

```
# The box plot
plt.figure(figsize=(10,6))
sns.boxplot(x= 'countrycode', y= 'log_ind_prod_sa', data=df_dataset)
plt.title('Production distribution per country')
plt.xlabel('Country with country code')
plt.ylabel('Production capabilities')
plt.show()
```




```
# Normality test to check the curve only in plot
plt.figure(figsize=(10,6))
sns.histplot(df_dataset['ind_prod_const_sa'], kde=True, bins=30)
plt.title('Production distribution')
plt.xlabel('Production capabilities')
plt.ylabel('Frequency')
plt.show()
```



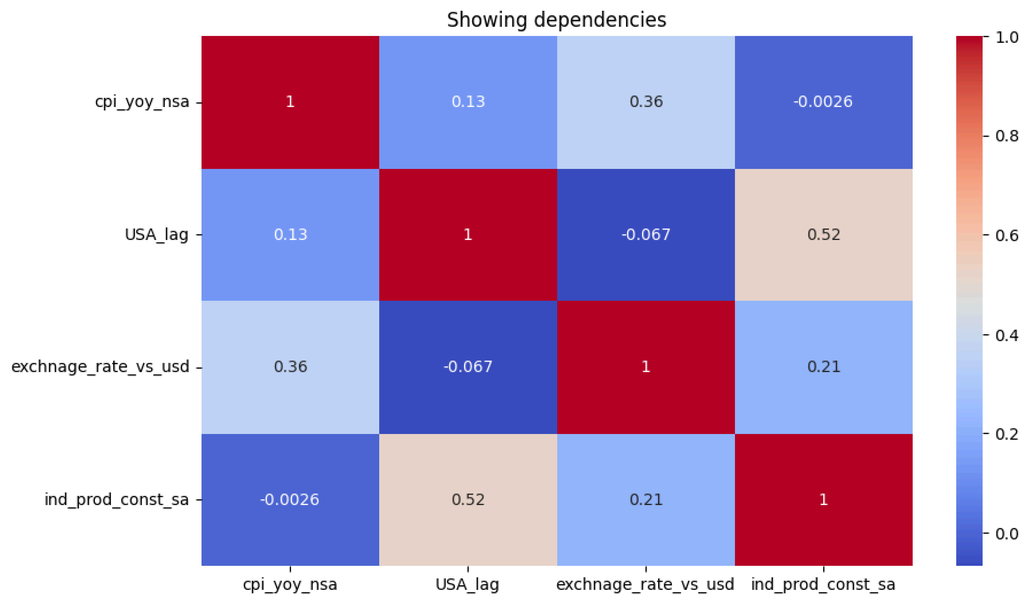

```
# Normality test to check the curve only in plot
plt.figure(figsize=(10,6))
sns.histplot(df_dataset['log_ind_prod_sa'], kde=True, bins=30)
plt.title('Production distribution')
plt.xlabel('Production capabilities')
plt.ylabel('Frequency')
plt.show()
```




```
#checking for correlation
correlation_f= df_dataset[['cpi_yoy_nsa','USA_lag','exchnage_rate_vs_usd','ind_prod_const_sa']].corr()
correlation_f
```

	cpi_yoy_nsa	USA_lag	exchnage_rate_vs_usd	ind_prod_const_sa
cpi_yoy_nsa	1.000000	0.129464	0.357678	-0.002609
USA_lag	0.129464	1.000000	-0.067300	0.523040
exchnage_rate_vs_usd	0.357678	-0.067300	1.000000	0.213621
ind_prod_const_sa	-0.002609	0.523040	0.213621	1.000000

```
#heatmap
plt.figure(figsize=(10,6))
sns.heatmap(correlation_f, annot= True, cmap='coolwarm')
plt.title('Showing dependencies')
plt.show()
```

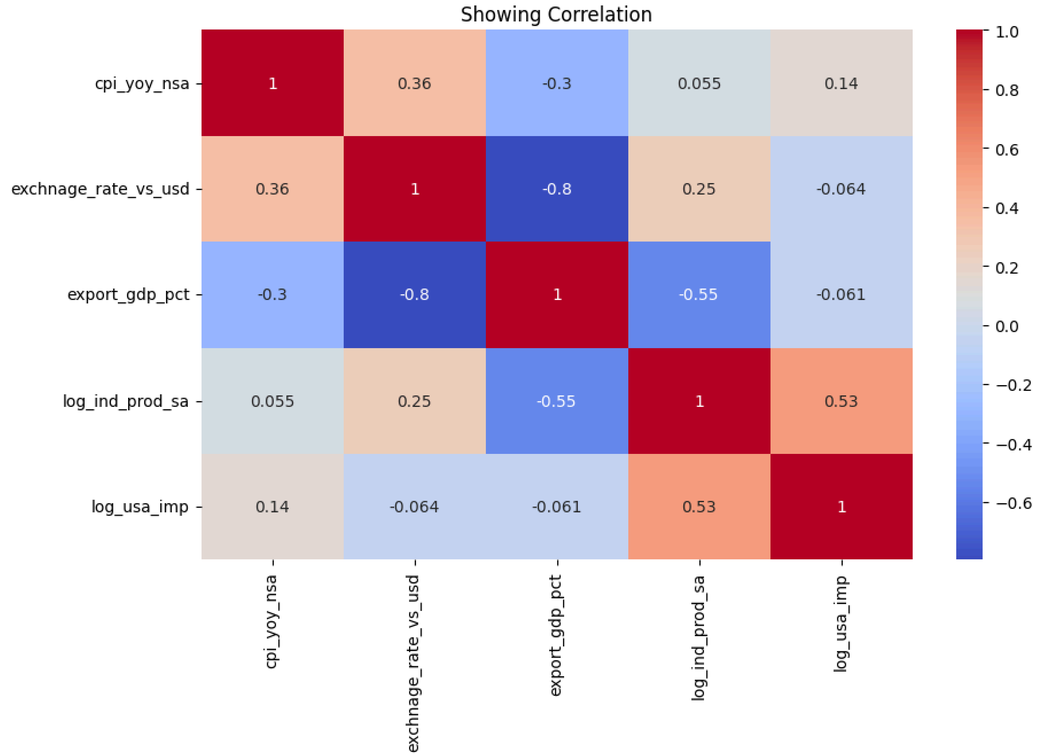



```
# Creating correlation log variables to check the changes and how much it is affected post filtering all outliers
```

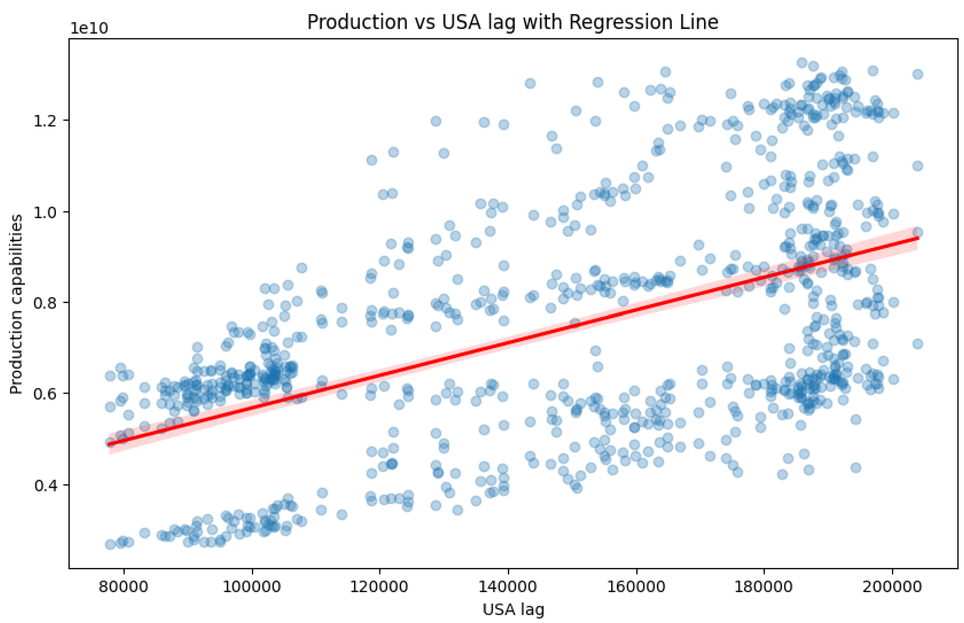
```
correlation_log= df_dataset[['cpi_yoy_nsa','exchnage_rate_vs_usd', 'export_gdp_pct', 'log_ind_prod_sa', 'log_usa_imp']].corr()
correlation_log
```

	cpi_yoy_nsa	exchnage_rate_vs_usd	export_gdp_pct	log_ind_prod_sa	log_usa_imp
cpi_yoy_nsa	1.000000	0.357678	-0.303041	0.055253	0.141590
exchnage_rate_vs_usd	0.357678	1.000000	-0.796807	0.251158	-0.063743
export_gdp_pct	-0.303041	-0.796807	1.000000	-0.549360	-0.060578
log_ind_prod_sa	0.055253	0.251158	-0.549360	1.000000	0.527976
log_usa_imp	0.141590	-0.063743	-0.060578	0.527976	1.000000

```
# creating heatmap
plt.figure(figsize=(10,6))
sns.heatmap(correlation_log, annot= True, cmap='coolwarm')
plt.title('Showing Correlation')
plt.show()
```

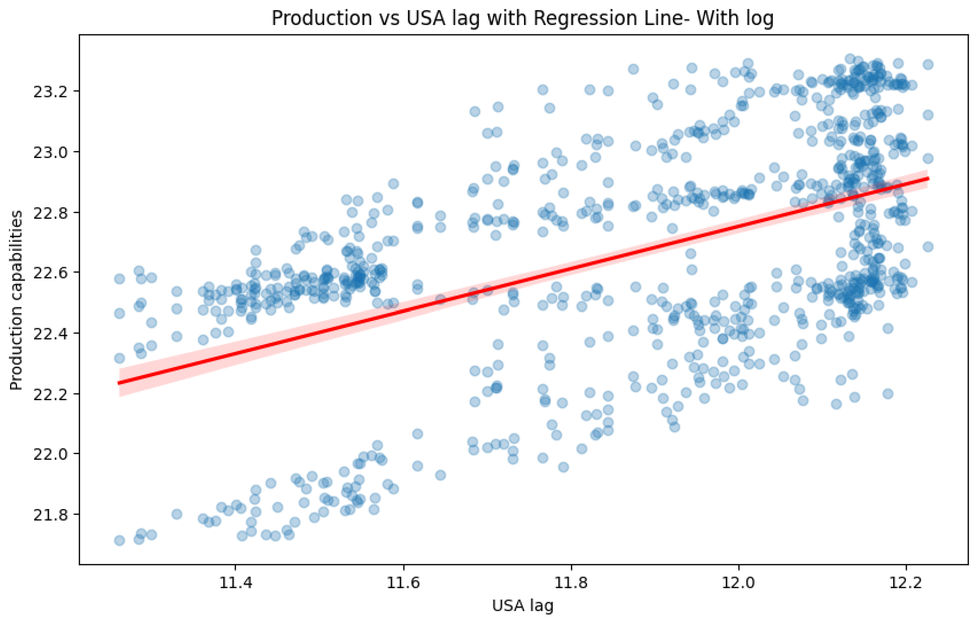



```
# Creating a scatter plot to show the relationship between the production capacity of asia vs the USA lag

plt.figure(figsize=(10,6))
sns.regplot(x= 'USA_lag', y='ind_prod_const_sa', data= df_dataset, scatter_kws={'alpha':0.3}, line_kws={'color': 'red'})
plt.title('Production vs USA lag with Regression Line')
plt.xlabel('USA lag')
plt.ylabel('Production capabilities')
plt.show()
```




```
# Creating a scatter plot to show the relationship between the production capacity of asia vs the USA lag post using log

plt.figure(figsize=(10,6))
sns.regplot(x= 'log_usa_imp', y='log_ind_prod_sa', data= df_dataset, scatter_kws={'alpha':0.3}, line_kws={'color': 'red'})
```

```
plt.title('Production vs USA lag with Regression Line- With log')
plt.xlabel('USA lag')
plt.ylabel('Production capabilities')
plt.show()
```



Hypothesis Testing: The objective is clear for the assignment, we need to analyze whether the production scenario increases in Asian market when there is an excessive need or demand in US economy.

Null Hypothesis, *Ho* : There is no relation between the two economies neiher of the scenario affects the market. Alternate Hypothesis, *H1* : States that yes there is a significant relation between the two economies as if the demand increases in US economy the production capacity will increase in Asian market as they need to produce more to export.

Here we will be using two variables: Dependent variable(Y) will be Asian Market Industrial production index Seasonally adjusted ~ ***'ind_prod_const_sa'*** Independent variable is US Imports which is lagged by a month of time as shipping is a factor here ~ ***USA_lag***

```
# Create country dummies
country_dummies = pd.get_dummies(df_dataset['countrycode'], drop_first=True).astype(int)
```

```
#Creating a global value chain to check which country is more sensitive to US demands
df_dataset['gvc_integration']= df_dataset['log_usa_imp'] * df_dataset['export_gdp_pct']
```

```
# combining log variable, dummie and interaction
X= pd.concat([df_dataset[['log_usa_imp','gvc_integration','cpi_yoy_nsa','exchnage_rate_vs_usd']], country_dummies], axis=1)
```

```
# Hypothesis testing
import statsmodels.formula.api as sm
import statsmodels.api as sm_api # Import statsmodels.api
```

```
X = sm_api.add_constant(X)
y = df_dataset['log_ind_prod_sa']
X = X.apply(pd.to_numeric, errors='coerce')
```

```
# run the model
model = sm_api.OLS(endog=y, exog=X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:        log_ind_prod_sa   R-squared:                       0.866
Model:                            OLS   Adj. R-squared:                  0.864
Method:                 Least Squares   F-statistic:                     827.4
Date:                Tue, 17 Mar 2026   Prob (F-statistic):               0.00
Time:                        22:05:27   Log-Likelihood:                 521.62
No. Observations:                 908   AIC:                            -1027.
Df Residuals:                     900   BIC:                            -988.8
Df Model:                           7
```

```
Covariance Type:            nonrobust
==============================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                     14.4459      0.229     63.134      0.000      13.997      14.895
log_usa_imp                0.7061      0.018     38.377      0.000       0.670       0.742
gvc_integration        -2.949e-06   3.08e-05     -0.096      0.924   -6.35e-05    5.76e-05
cpi_yoy_nsa               -0.0063      0.003     -2.494      0.013      -0.011      -0.001
exchnage_rate_vs_usd      -0.0003      0.002     -0.221      0.825      -0.003       0.003
PHL                       -0.2054      0.070     -2.937      0.003      -0.343      -0.068
SGP                       -0.5667      0.040    -14.212      0.000      -0.645      -0.488
THA                        0.2074      0.052      4.007      0.000       0.106       0.309
==============================================================================
Omnibus:                        2.173   Durbin-Watson:                   2.064
Prob(Omnibus):                  0.337   Jarque-Bera (JB):                2.059
Skew:                           0.076   Prob(JB):                        0.357
Kurtosis:                       3.176   Cond. No.                     7.04e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 7.04e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Double-click (or enter) to edit

```
# Using Spearman and pearson correlational check
import scipy.stats as stats
```

```
Start coding or generate with AI.
```

```
df_data_clean= df_dataset[['log_ind_prod_sa', 'log_usa_imp']].dropna()
```

Double-click (or enter) to edit

```
r_coeff, p_val= stats.pearsonr(df_data_clean['log_ind_prod_sa'], df_data_clean['log_usa_imp'])
print(f'Pearson Correlation Coefficient: {r_coeff}')
print(f'P-value: {p_val}')
```

```
Pearson Correlation Coefficient: 0.5279763283621385
P-value: 2.5694267093511764e-66
```

```
# using spearmanr
```

```
r_coeffS, p_vals= stats.spearmanr(df_data_clean['log_ind_prod_sa'], df_data_clean['log_usa_imp'])
print(f'Spearman Correlation Coefficient: {r_coeffS}')
print(f'P-value: {p_vals}')
```

```
Spearman Correlation Coefficient: 0.5055712041595554
P-value: 4.42366512966866e-60
```

```
# Linear regression
```

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

```
X = pd.concat([df_dataset[['log_usa_imp','gvc_integration','cpi_yoy_nsa','exchnage_rate_vs_usd']], country_dummies], axis=1)
X = sm_api.add_constant(X)
y= df_dataset['log_ind_prod_sa']

X_train, X_test, y_train, y_test= train_test_split(X,y, test_size= 0.2, random_state= 42)

lin_reg= LinearRegression()
lin_reg.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
# Testing the model
y_pred= lin_reg.predict(X_test)
```

```
#calculating r2_score
```

```
r_score= r2_score(y_test, y_pred)
print(f'R-squared Score: {r_score}')
```

```
R-squared Score: 0.8977224017278848
```

```
# Training and testing for double check
train_score= lin_reg.score(X_train, y_train)
test_score= lin_reg.score(X_test, y_test)
print(f'Training Score: {train_score}')
print(f'Testing Score: {test_score}')
```

```
Training Score: 0.8565309066759728
Testing Score: 0.8977224017278848
```

```
X = df_dataset[['log_usa_imp']]
y= df_dataset[['log_ind_prod_sa']]

X_train, X_test, y_train, y_test= train_test_split(X,y, test_size= 0.2, random_state= 42)

lin_reg= LinearRegression()
lin_reg.fit(X_train, y_train)
```

```
LinearRegression
LinearRegression()
```

```
y_pred= lin_reg.predict(X_test)
R_square= r2_score(y_test, y_pred)
print(f'Testing score: {R_square}')
```

```
Testing score: 0.30380426214398293
```

Start coding or generate with AI.

Double-click (or enter) to edit